A

Project Report

on

WriteMate: Smart Assistive Writing System

submitted as partial fulfillment for the award of

BACHELOR OF TECHNOLOGY

DEGREE

SESSION 2025-26

in

Computer Science & Engineering

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(Formerly UPTU) **May, 2026**

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgement has been made in the text.

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CERTIFICATE

This is to certify that the Project Report entitled "WriteMate: A Multimodal Assistive Writing System for Individuals with Physical Disabilities" which is submitted by Abhay Kauts, Aditya Chandra Verma, Dhairya Gupta, Arnav Chaudhary in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Computer Science & Engineering of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates' own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B. Tech Final Year. We owe a special debt of gratitude to Pranay Meshram, Department of Computer Science & Engineering, KIET Group of Institutions, Ghaziabad, for his constant support and guidance throughout the course of our work. His sincerity, thoroughness, and constructive feedback have been a constant source of inspiration for us.

We also take this opportunity to acknowledge the contribution of Dr. Vineet Sharma, Head of the Department of Computer Science & Engineering, KIET Group of Institutions, Ghaziabad, for his full support and assistance during the development of the project. We are also thankful to the faculty members of the department for their guidance, cooperation, and encouragement.

We acknowledge our friends, classmates, and all individuals who provided feedback during prototype testing. Their observations helped us refine the usability, reliability, and practical relevance of WriteMate.

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ABSTRACT

WriteMate is a multimodal assistive writing system developed to help individuals with physical disabilities perform handwritten tasks independently without relying on human scribes. The system accepts input through spoken commands and Indian Sign Language gestures, processes the input locally on a Raspberry Pi 5, and converts the recognized text into physical handwritten output using a CNC-based writing mechanism.

The project addresses practical limitations of scribe-based writing, including dependence on availability, privacy concerns, communication errors, and inconsistent handwriting. WriteMate uses lightweight speech recognition, MediaPipe-based hand landmark extraction, a fine-tuned gesture recognition model, and G-code generation to translate user intent into controlled pen motion. The device is designed for offline use so that it remains dependable in examination halls, hospitals, administrative offices, and areas where internet connectivity is limited.

Prototype evaluation demonstrates a mean letter-recognition accuracy of 92.7 percent across the English alphabet, a macro F1 score of 0.924, and end-to-end latency below 150 ms. CNC calibration produced consistent and legible handwriting across extended sessions. The system therefore provides a practical, affordable, and privacy-preserving alternative to manual scribe-based assistance.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ASR	Automatic Speech Recognition
CNC	Computer Numerical Control
CNN	Convolutional Neural Network
FPS	Frames Per Second
GPIO	General Purpose Input Output
ISL	Indian Sign Language
ML	Machine Learning
VAD	Voice Activity Detection
WER	Word Error Rate
YOLO	You Only Look Once

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

This section introduces the need for assistive writing as a personal requirement rather than just a technical specification. We explore why handwritten work remains essential for people with physical disabilities. While many individuals can think, speak, and compose their thoughts independently, they are often forced to rely on a second person just to perform the physical task of writing.

We also establish why a multimodal interface is so important. Some users feel more comfortable speaking, while others rely on sign language, and many may need to switch between both depending on where they are. By offering multiple input methods, the system becomes far more inclusive and adaptable to the user's life.

WriteMate treats assistive writing as a design commitment to the user. Our goal is to create a device that translates a person's intent into physical writing while ensuring they stay fully in charge of their content, their speed, and how they choose to communicate.

Finally, this section defines the project's focus. We aren't trying to build a complex general-purpose robot or a digital assistant; instead, we are creating a dedicated writing partner that can be measured by how accurately it recognizes input and how well it produces legible, real-world handwriting.

For need for assistive writing, the most important design question is whether the proposed system can be understood by a user without technical training. WriteMate therefore avoids unnecessary interaction steps: the user selects an input mode, provides speech or gesture input, and receives handwritten output after validation.

The project also considers the institutional setting in which the device may be used. In an examination hall, a student cannot spend time configuring complex software; in a hospital, the device should support quick dictated notes or form entries; in an office, privacy and repeatable writing quality become important.

The introduction of this topic also helps explain why WriteMate is more than a typing substitute. Typing assistance creates digital text, but many formal processes still need visible writing on paper. The project is designed around that practical requirement.

From a documentation perspective, Section 1.1 helps show that the project decisions are not arbitrary. The treatment of need for assistive writing follows from the problem statement and leads naturally into the design, testing, or conclusion that appears later in the report.

The connection between handwritten examinations and privacy during dictated writing is especially useful because it shows the movement from input requirement to output requirement. WriteMate begins with user expression and ends with handwriting, so both ends of the pipeline must be discussed.

By including official forms and medical records, the section also records the intermediate engineering decisions. These decisions make the prototype more reliable than a simple direct connection between a recognizer and a plotter.

1.1 INTRODUCTION

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1.2 PROBLEM STATEMENT: THE BARRIER TO INDEPENDENT WRITING

For individuals with physical disabilities, the reliance on a human scribe for handwriting introduces significant and unnecessary barriers to independence. This section outlines the core issues WriteMate addresses by creating a practical bridge across a critical gap in current accessibility solutions.

Depending on a scribe compromises the user's agency. It creates problems with availability (scribes may not always be present), communication errors, unpredictable variations in writing speed and quality, and, critically, a complete loss of content privacy, especially in formal settings like exams or when handling personal medical notes.

To truly restore autonomy, the assistive system must be entirely independent. Therefore, WriteMate is designed with strict, measurable targets: it must be affordable, operate completely offline for maximum reliability, and achieve a high degree of recognition accuracy. This disciplined approach elevates WriteMate beyond a mere technical demonstration; success is defined by its ability to reliably produce legible written output that can be read, checked, and accepted in any formal setting.

This motivation dictates our entire project workflow—from the literature review to the methodology and final results. We evaluate the system not just on recognition metrics, but on real-world outcomes like latency, handwriting quality, and robustness, ensuring that the same practical problems described here are revisited and overcome in the testing phase.

Accessibility must be foundational, not an afterthought. The system's design—its multimodal input, offline engine, and precise motion control—is completely centered around the user. WriteMate converts a person's intent into physical writing, allowing the user to remain fully in control of the content, the pace, and the preferred input method, fostering independence, predictability, and confidence.

1.3 RATIONALE

Our motivation for WriteMate goes beyond technical specifications; it centers on a deep, user-driven requirement. We aim to solve a fundamental injustice: the fact that many individuals can think, speak, sign, and compose their thoughts independently, yet remain completely dependent on another person—a scribe—for the simple, physical act of writing.

This project is made possible by the intersection of modern embedded AI and accessible mechatronics. We chose components like the Raspberry Pi 5, lightweight speech recognition models, hand landmark extraction, and precise stepper-motor control not just for speed, but for cost and reliability. Each technical choice directly supports the user experience: the Pi provides the local processing power needed for comfort, and the precise motors ensure the user can trust the system to produce consistent, legible output over repeated sessions.

We also intentionally developed a multimodal interface. Real life is complicated; some users may prefer speech in a quiet room, while others require sign language in a noisy environment. By offering both input modes, the system becomes far more inclusive and adaptable to the user's personal needs and environment.

Crucially, we defined the project's boundaries very strictly. WriteMate is not an attempt to build a complex general-purpose robot or a digital assistant. Instead, it is a highly focused assistive writing partner. Its success is measured by practical, user-centered standards: how accurately it recognizes input, its response speed (latency), the real-world quality of the handwriting it produces, and overall usability.

The most important design principle was simplicity. WriteMate avoids unnecessary complexity so that a user without technical training can operate it immediately. Whether in a high-stakes setting like an examination hall, where a student cannot waste time configuring software, or in a hospital or office, where privacy and consistent form-filling are essential, the device must just work. This focus on institutional settings is why we absolutely mandate offline operation; examinations, rural areas, and privacy-sensitive workflows cannot depend on unreliable cloud services.

Ultimately, our approach to WriteMate is an act of design responsibility. The entire device is constructed around restoring a person's autonomy. It converts their intent into written output, allowing them to remain fully in charge of their content, their pace, and their preferred method of communication. By narrowing the workflow to this single, clear

task—converting accessible input into readable handwritten output—we ensure that WriteMate contributes directly to independent writing, reduces examination stress, guarantees private communication, and helps build inclusive infrastructure.

1.4 OBJECTIVES

In the context of WriteMate, project objectives explains why the project was framed as a complete assistive workflow. The objective is not to replace communication ability; it is to remove the mechanical barrier between a user's intended content and the written page.

The project team treated multimodal input and sub-150 ms response as immediate user concerns, while offline recognition and legible physical output were handled as engineering constraints. This separation helped the design remain practical without losing sight of the people for whom the system is being built.

The introduction further clarifies why cloud dependence is avoided. Network-based tools may work in ordinary conditions, but examinations, rural institutions, hospitals, and privacy-sensitive workflows require a device that can function locally.

In Section 1.4, this point is specifically applied to project objectives.

WriteMate therefore approaches project objectives as a design responsibility. The proposed device must convert intention into written output while allowing the user to remain in control of the content, pace, and input mode.

This section also helps define the boundary of the project. The goal is not to build a general-purpose robot or a full natural-language assistant, but to create a focused assistive writing system that can be evaluated through recognition accuracy, latency, handwriting quality, and usability.

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The treatment of project objectives follows from the problem statement and leads naturally into the design, testing, or conclusion that appears later in the report.

The connection between multimodal input and legible physical output is especially useful because it shows the movement from input requirement to output requirement.

WriteMate begins with user expression and ends with handwriting, so both ends of the pipeline must be discussed.

By including sub-150 ms response and offline recognition, the section also records the intermediate engineering decisions.

These decisions make the prototype more reliable than a simple direct connection between a recognizer and a plotter.

1.5 PROJECT DESCRIPTION

who must produce written content without depending on a scribe. This use case helped avoid unnecessary features and kept the design focused.

The project scope deliberately remains focused on handwritten output. Many digital tools already support typing or screen display, but WriteMate targets the still-common requirement of producing marks on paper.

WriteMate therefore approaches overall project description as a design responsibility. The proposed device must convert intention into written output while allowing the user to remain in control of the content, pace, and input mode.

This section also helps define the boundary of the project. The goal is not to build a general-purpose robot or a full natural-language assistant, but to create a focused assistive writing system that can be evaluated through recognition accuracy, latency, handwriting quality, and usability. In Section 1.5, this point is specifically applied to overall project description.

The reasoning behind overall project description also reflects the Indian educational and administrative context. Paper-based processes remain common, and support for disabled users often depends on manual arrangements rather than reliable assistive infrastructure.

A device such as WriteMate can reduce some of this dependence by providing the same handwriting mechanism each time. This improves consistency and reduces the uncertainty that comes from arranging scribes at short notice.

The design attention given to input acquisition, model inference, text buffer management, and G-code execution supports both fairness and dignity. Fairness is improved when the writing process is predictable; dignity is improved when the user does not have to reveal private content unnecessarily.

This section prepares the reader for the later chapters by showing that the technical system is motivated by a concrete social need rather than by novelty alone.

CHAPTER 2
LITERATURE REVIEW

2.1 OVERVIEW

In reviewing scope of literature review, the report focuses on methods that are realistic for embedded hardware. Accuracy is important, but latency, model size, privacy, and hardware cost are equally relevant for the proposed system.

Several systems perform well under narrow assumptions. WriteMate draws from those results but expands the design around offline ASR, ISL recognition, edge machine learning, and automated plotters, because a user-facing assistive device must work across input, processing, and output stages.

By comparing related work, the report avoids presenting WriteMate as an isolated invention. It is better understood as a practical integration of existing research areas around a specific accessibility need.

For this reason, the literature review is organized around the actual project pipeline. Speech recognition, gesture recognition, assistive systems, embedded inference, and CNC plotting are examined as connected layers that must operate together.

The outcome of the review is a set of design expectations: the system should use lightweight models, validate uncertain input, avoid cloud dependence, and translate recognized text into motion commands with sufficient mechanical precision.

The literature on scope of literature review also helps select realistic performance expectations. A low-cost assistive system cannot be judged by the standards of large cloud platforms or industrial machines, but it must still be responsive and accurate enough for use.

The report therefore interprets related work through constraints such as Raspberry Pi processing, camera quality, microphone noise, motor precision, and paper handling. These constraints match the actual prototype environment.

The attention given to offline ASR and automated plotters is especially useful because they represent opposite ends of the pipeline: input interpretation and final output quality.

This balanced review prevents the project from becoming one-sided. It gives equal importance to software intelligence and physical execution.

Section 2.1 also creates traceability between the report objective and the prototype implementation. In the case of scope of literature review, the discussion connects offline ASR with the early design requirement, ISL recognition with the processing stage, and edge machine learning with the reliability expected from the final output.

A practical reader should be able to follow how this section affects the system that was built. The explanation of scope of literature review is therefore linked with the microphone, camera, Raspberry Pi, recognition models, buffer logic, and CNC writer wherever those parts are relevant.

2.2 SPEECH RECOGNITION

The study of offline speech recognition helps locate WriteMate among speech systems, sign-language systems, edge-AI prototypes, and CNC writing machines. Each area solves part of the problem, but the integration gap remains important.

The literature suggests a useful direction but not a complete ready-made solution. Techniques involving privacy and limited connectivity can be adopted, while issues related to restricted vocabulary and noise reduction require additional integration work.

The chapter therefore prepares the methodology by identifying which ideas can be reused and which problems must be solved within the project itself.

For this reason, the literature review is organized around the actual project pipeline. Speech recognition, gesture recognition, assistive systems, embedded inference, and CNC plotting are examined as connected layers that must operate together. In Section 2.2, this point is specifically applied to offline speech recognition.

The outcome of the review is a set of design expectations: the system should use lightweight models, validate uncertain input, avoid cloud dependence, and translate recognized text into motion commands with sufficient mechanical precision. In Section 2.2, this point is specifically applied to offline speech recognition.

In comparing earlier systems, the report pays attention to what each one leaves unresolved. Some support speech but not sign input, some recognize gestures but stop at digital text, and some write physically but depend on a single input source.

This makes offline speech recognition important for identifying the exact contribution of WriteMate. The project does not claim to invent speech recognition, gesture recognition, or CNC plotting from scratch; it contributes through purposeful integration.

The role of privacy, limited connectivity, restricted vocabulary, and noise reduction is to show how separate research lines can be combined around one accessibility requirement. This is where the report's literature review becomes directly connected to the project design.

The result is a research gap that is specific and defensible: a portable offline device that accepts voice and ISL input and produces physical handwriting.

The same reasoning applies to restricted vocabulary and noise reduction. They influence how the prototype behaves during repeated use, where consistency matters as much as a single successful demonstration.

The literature on speech preprocessing is useful because it identifies proven tools as well as limitations in earlier work. WriteMate uses these observations to combine methods rather than duplicate a single existing system.

The main lesson from this review is that component-level performance must be connected to system-level usability. A classifier with good accuracy is valuable only when its output can be converted into stable, readable writing.

The review supports the conclusion that WriteMate's novelty lies in the complete offline multimodal-to-handwriting workflow rather than in a single standalone algorithm.

For this reason, the literature review is organized around the actual project pipeline. Speech recognition, gesture recognition, assistive systems, embedded inference, and CNC plotting are examined as connected layers that must operate together. In Section 2.2, this point is specifically applied to speech preprocessing.

Another lesson from the reviewed work is that accessibility systems must be evaluated with the user's task in mind. For WriteMate, the task is not merely recognizing an alphabet; it is completing a writing action that would otherwise require a scribe.

That distinction changes how speech preprocessing is interpreted. Accuracy, latency, and precision are not abstract metrics; they represent whether a user can produce a page confidently and within a reasonable time.

These insights form the bridge between the review chapter and the proposed methodology.

The details involving sampling rate, voice activity detection, spectral filtering, and token validation help convert the project from an idea into a usable prototype. They show that the team considered both software behaviour and physical constraints.

2.3 GESTURE RECOGNITION

Work on isl gesture recognition provides the technical background for choosing the project architecture. Prior systems demonstrate useful methods, but they also reveal why a device that produces physical handwriting needs more than isolated recognition accuracy.

Among the recurring ideas in the literature are hand shape, orientation, landmark extraction, and class ambiguity. For WriteMate, these ideas are filtered through the requirement that the final output must be physical handwriting, not only digital text.

This review also justifies the use of Raspberry Pi as an edge platform. It is powerful enough for lightweight recognition while still affordable and suitable for an academic prototype.

For this reason, the literature review is organized around the actual project pipeline. Speech recognition, gesture recognition, assistive systems, embedded inference, and CNC plotting are examined as connected layers that must operate together. In Section 2.3, this point is specifically applied to iSL gesture recognition.

The outcome of the review is a set of design expectations: the system should use lightweight models, validate uncertain input, avoid cloud dependence, and translate recognized text into motion commands with sufficient mechanical precision. In Section 2.3, this point is specifically applied to iSL gesture recognition.

WriteMate responds to this gap by combining preprocessing, restricted vocabulary, gesture stabilization, and mechanical calibration. These steps are less glamorous than model selection but are necessary for practical use.

The points hand shape, orientation, landmark extraction, and class ambiguity show the range of issues covered in the literature. Together they explain why the project cannot depend on a single algorithmic improvement.

The literature therefore supports the project structure: input processing, recognition, command preparation, and CNC execution are treated as linked parts of one system.

Section 2.3 also creates traceability between the report objective and the prototype implementation. In the case of iSL gesture recognition, the discussion connects hand shape with the early design requirement, orientation with the processing stage, and landmark extraction with the reliability expected from the final output.

A practical reader should be able to follow how this section affects the system that was built. The explanation of iSL gesture recognition is therefore linked with the microphone, camera,

Raspberry Pi, recognition models, buffer logic, and CNC writer wherever those parts are relevant.

The section further avoids treating accessibility as a vague idea. It translates the need for class ambiguity into concrete design behaviour: the system should respond predictably, preserve user control, and create written output that can be checked on paper.

This review section examines mediapipe and deep models from the perspective of assistive writing. The main question is whether existing techniques can support an offline, portable, multimodal system rather than a laboratory-only demonstration.

For this reason, the literature review is organized around the actual project pipeline. Speech recognition, gesture recognition, assistive systems, embedded inference, and CNC plotting are examined as connected layers that must operate together. In Section 2.3, this point is specifically applied to mediaPipe and deep models.

The outcome of the review is a set of design expectations: the system should use lightweight models, validate uncertain input, avoid cloud dependence, and translate recognized text into motion commands with sufficient mechanical precision. In Section 2.3, this point is specifically applied to mediaPipe and deep models.

The review of mediaPipe and deep models also shows why offline operation is repeatedly emphasized. Cloud systems can be accurate, but they introduce dependence on connectivity and external data handling, both of which are problematic in examinations and private documentation.

Embedded approaches may require more careful optimization, yet they give the user and institution greater control. This trade-off is acceptable for WriteMate because the vocabulary and output task are focused.

Research involving 21 hand landmarks and CNN features provides useful evidence that local processing is feasible. Work related to YOLO-based classification and temporal stability further supports the mechanical side of the project.

These observations guide the methodology chapter, where the selected tools are adapted into a working prototype rather than discussed only in theory.

Another important aspect of Section 2.3 is its connection with error prevention. For mediaPipe and deep models, the system must reduce mistakes before they reach the writing

stage, because errors written on paper are harder to ignore than errors displayed temporarily on a screen.

This is why the report repeatedly connects 21 hand landmarks and CNN features with validation. The device is expected to filter unclear input, stabilize uncertain gestures, and organize recognized text before G-code is generated.

The same reasoning applies to YOLO-based classification and temporal stability. They influence how the prototype behaves during repeated use, where consistency matters as much as a single successful demonstration.

2.4 ASSISTIVE TECHNOLOGIES

The literature related to existing assistive tools shows how existing research contributes one part of the WriteMate pipeline. This section reviews the area with attention to discussing why many accessibility tools stop at digital output, because the project depends on both recognition reliability and practical deployment.

The reviewed works commonly discuss screen readers, voice assistants, speech-to-text editors, and digital-only interfaces. These themes support the design choices made in WriteMate, especially the decision to keep inference local and to validate recognized tokens before passing them to the writing mechanism.

The research gap becomes clear when the reviewed systems are compared. Most solutions either accept only one input modality or produce only digital output, while WriteMate joins voice, ISL gestures, offline processing, and CNC handwriting.

The outcome of the review is a set of design expectations: the system should use lightweight models, validate uncertain input, avoid cloud dependence, and translate recognized text into motion commands with sufficient mechanical precision. In Section 2.4, this point is specifically applied to existing assistive tools.

In the literature, existing assistive tools is often discussed as an independent research problem. WriteMate uses the same research area differently: it evaluates whether the method can serve a complete assistive writing workflow with physical output.

Several prior systems achieve useful performance in recognition or plotting, but the reviewed work shows that isolated performance does not automatically become a usable assistive device. Integration remains the central challenge.

The relevance of screen readers and voice assistants is therefore judged by how well they support later stages such as validation, buffering, and motion control. The project adopts only those ideas that fit the offline embedded setting.

This review approach keeps the literature chapter focused. Instead of listing unrelated papers, it explains how each research area contributes to the final WriteMate architecture.

Another important aspect of Section 2.4 is its connection with error prevention. For existing assistive tools, the system must reduce mistakes before they reach the writing stage, because errors written on paper are harder to ignore than errors displayed temporarily on a screen.

The main lesson from this review is that component-level performance must be connected to system-level usability. A classifier with good accuracy is valuable only when its output can be converted into stable, readable writing. In Section 2.4, this point is specifically applied to assistive handwriting systems.

The review supports the conclusion that WriteMate's novelty lies in the complete offline multimodal-to-handwriting workflow rather than in a single standalone algorithm. In Section 2.4, this point is specifically applied to assistive handwriting systems.

For this reason, the literature review is organized around the actual project pipeline. Speech recognition, gesture recognition, assistive systems, embedded inference, and CNC plotting are examined as connected layers that must operate together. In Section 2.4, this point is specifically applied to assistive handwriting systems.

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That distinction changes how assistive handwriting systems is interpreted. Accuracy, latency, and precision are not abstract metrics; they represent whether a user can produce a page confidently and within a reasonable time.

The literature connected with internet dependence and limited evaluation supports the use of lightweight models, while work on single input modality and physical output supports the need for careful data and motion design.

Section 2.4 also creates traceability between the report objective and the prototype implementation. In the case of assistive handwriting systems, the discussion connects single input modality with the early design requirement, internet dependence with the processing stage, and limited evaluation with the reliability expected from the final output.

A practical reader should be able to follow how this section affects the system that was built. The explanation of assistive handwriting systems is therefore linked with the microphone, camera, Raspberry Pi, recognition models, buffer logic, and CNC writer wherever those parts are relevant.

The section further avoids treating accessibility as a vague idea. It translates the need for physical output into concrete design behaviour: the system should respond predictably, preserve user control, and create written output that can be checked on paper.

2.5 EMBEDDED SYSTEMS

The literature related to embedded machine learning shows how existing research contributes one part of the WriteMate pipeline. This section reviews the area with attention to showing why edge execution is necessary for privacy and responsiveness, because the project depends on both recognition reliability and practical deployment.

The reviewed works commonly discuss model optimization, memory limits, ARM processing, and real-time inference. These themes support the design choices made in WriteMate, especially the decision to keep inference local and to validate recognized tokens before passing them to the writing mechanism.

The research gap becomes clear when the reviewed systems are compared. Most solutions either accept only one input modality or produce only digital output, while WriteMate joins voice, ISL gestures, offline processing, and CNC handwriting. In Section 2.5, this point is specifically applied to embedded machine learning.

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In the literature, embedded machine learning is often discussed as an independent research problem. WriteMate uses the same research area differently: it evaluates whether the method can serve a complete assistive writing workflow with physical output.

Section 2.5 also creates traceability between the report objective and the prototype implementation. In the case of embedded machine learning, the discussion connects model optimization with the early design requirement, memory limits with the processing stage, and ARM processing with the reliability expected from the final output.

2.6 CNC WRITING SOLUTIONS

This review section examines cnc plotter systems from the perspective of assistive writing. The main question is whether existing techniques can support an offline, portable, multimodal system rather than a laboratory-only demonstration.

Existing studies show that X-Y movement and NEMA 17 motors are technically achievable, but practical use also depends on A4988 drivers and servo pen lift. This is where many previous systems become incomplete for the specific problem of scribe replacement.

The literature does not remove the need for calibration and testing. Even if recognition models perform well, the writing mechanism introduces mechanical sources of error that must be measured separately. In Section 2.6, this point is specifically applied to cNC plotter systems.

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The outcome of the review is a set of design expectations: the system should use lightweight models, validate uncertain input, avoid cloud dependence, and translate recognized text into motion commands with sufficient mechanical precision. In Section 2.6, this point is specifically applied to cNC plotter systems.

The review of cNC plotter systems also shows why offline operation is repeatedly emphasized. Cloud systems can be accurate, but they introduce dependence on connectivity and external data handling, both of which are problematic in examinations and private documentation.

Embedded approaches may require more careful optimization, yet they give the user and institution greater control. This trade-off is acceptable for WriteMate because the vocabulary and output task are focused. In Section 2.6, this point is specifically applied to cNC plotter systems.

Research involving X-Y movement and NEMA 17 motors provides useful evidence that local processing is feasible. Work related to A4988 drivers and servo pen lift further supports the mechanical side of the project.

These observations guide the methodology chapter, where the selected tools are adapted into a working prototype rather than discussed only in theory. In Section 2.6, this point is The details involving X-Y movement, NEMA 17 motors, A4988 drivers, and servo pen lift help convert the project from an idea into a usable prototype. They show that the team considered both software behaviour and physical constraints.

2.7 RESEARCH GAP

This review section examines identified research gap from the perspective of assistive writing. The main question is whether existing techniques can support an offline, portable, multimodal system rather than a laboratory-only demonstration.

Existing studies show that dual input and offline operation are technically achievable, but practical use also depends on portable device and scribe replacement. This is where many previous systems become incomplete for the specific problem of scribe replacement.

The literature does not remove the need for calibration and testing. Even if recognition models perform well, the writing mechanism introduces mechanical sources of error that must be measured separately. For this reason, the literature review is organized around the actual project pipeline. Speech recognition, gesture recognition, assistive systems, embedded inference, and CNC plotting are examined as connected layers that must operate together. In Section 2.7, this point is specifically applied to identified research gap.

The outcome of the review is a set of design expectations: the system should use lightweight models, validate uncertain input, avoid cloud dependence, and translate recognized text into

motion commands with sufficient mechanical precision. In Section 2.7, this point is specifically applied to identified research gap.

The review of identified research gap also shows why offline operation is repeatedly emphasized. Cloud systems can be accurate, but they introduce dependence on connectivity and external data handling, both of which are problematic in examinations and private documentation..

Research involving dual input and offline operation provides useful evidence that local processing is feasible. Work related to portable device and scribe replacement further supports the mechanical side of the project.

These observations guide the methodology chapter, where the selected tools are adapted into a working prototype rather than discussed only in theory. The discussion in Section 2.7 can also be read as a design checkpoint. For this reason, the report gives attention to dual input, offline operation, portable device, and scribe replacement rather than presenting only final results. Each point contributes to the quality of the handwritten page.

This section therefore supports the credibility of the project. It gives enough detail for a reviewer to see how WriteMate was planned, built, and evaluated as a complete assistive writing system.

CHAPTER 3
PROPOSED METHODOLOGY

3.1 METHODOLOGY OVERVIEW

This section documents the construction of development methodology in enough detail to show how the prototype can be reproduced or improved. The focus is practical: data, processing, calibration, and integration are described as working steps.

Practical constraints shaped the implementation. The Raspberry Pi must process input quickly, the recognition modules must avoid unnecessary computation, and the CNC controller must receive commands that are already organized for writing.

By documenting this module carefully, the report makes the prototype understandable as a system rather than as a collection of unrelated components.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device.

A clear module boundary around development methodology makes later replacement and experimentation easier.

For example, the gesture classifier can be retrained with more samples, the speech implementation details.

This makes the project report useful not only as a record of completed work, but also as documentation for continuing development.

Another important aspect of Section 3.1 is its connection with error prevention, the system must reduce mistakes before they reach the writing stage, because errors written on paper are harder to ignore than errors displayed temporarily on a screen.

This is why the report repeatedly connects requirements and datasets with validation. The device is expected to filter unclear input, stabilize uncertain gestures, and organize recognized text before G-code is generated.

The same reasoning applies to model training and hardware integration. They influence how the prototype behaves during repeated use, where consistency matters as much as a single successful demonstration.

3.2 REQUIREMENT ANALYSIS

This part of the methodology explains how functional requirements was planned and integrated. The project depends on a chain of modules, so the design of one stage must account for the assumptions and limitations of the next.

In this module, voice mode defines the input condition, gesture mode affects processing reliability, writing output supports integration, and mode switching influences the final user experience.

Calibration is treated as a repeated process rather than a one-time setting. Pen pressure, motor movement, paper margins, and input confidence need periodic checking to preserve output quality.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.2, this point is specifically applied to functional requirements.

The methodology therefore supports both reliability and maintainability. If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. The methodology for functional requirements also reflects the limitations of low-cost hardware. Raspberry Pi 5 is capable, but it cannot be treated like an unlimited server; memory, processing time, and peripheral timing must be managed carefully.

To keep the system responsive, the recognition tasks are focused and the writing commands are structured. The project avoids unnecessary complexity that would increase latency without improving the user's writing experience.

The relationship between voice mode and gesture mode affects the input side, while writing output and mode switching influence how the system behaves after recognition. Both sides must remain synchronized.

This careful sequencing is what makes the prototype suitable for offline operation.

From a documentation perspective, Section 3.2 helps show that the project decisions are not arbitrary. The treatment of functional requirements follows from the problem statement and leads naturally into the design, testing, or conclusion that appears later in the report.

The connection between voice mode and mode switching is especially useful because it shows the movement from input requirement to output requirement. WriteMate begins with user expression and ends with handwriting, so both ends of the pipeline must be discussed.

Table 1

Requirement	Description
Voice input	Capture spoken letters and commands
Gesture input	Recognize ISL alphabet gestures
Offline operation	Avoid mandatory internet dependency
Physical output	Write recognized text on paper

The methodology for non-functional requirements describes how the project moves from concept to working prototype. This section focuses on defining quality targets for practical deployment, and it explains the practical decisions taken while building the WriteMate pipeline.

The implementation pays attention to latency, privacy, portability, and robustness. These points influence the module's behaviour under real conditions rather than only in ideal demonstrations.

This separation makes testing clearer, because an error can be traced to the speech module, gesture module, buffer, G-code generator, or motion system.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.2, this point is specifically applied to non-functional requirements.

The methodology therefore supports both reliability and maintainability. If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.2, this point is specifically applied to non-functional requirements.

The implementation of non-functional requirements was planned so that errors could be traced clearly. If recognition failed, the team could inspect the input and model stages; if handwriting failed, the G-code or mechanical calibration could be examined separately.

This separation is important because WriteMate combines different engineering domains. Audio processing, computer vision, embedded programming, and CNC control each have their own failure modes.Such modular development also supports future improvement. A better speech model or gesture classifier can be added without changing the entire writing mechanism.

the rest of the system becomes less dependable, because later modules assume that earlier decisions have produced clean and meaningful information.

For this reason, the report gives attention to latency, privacy, portability, and robustness rather than presenting only final results. Each point contributes to the quality of the handwritten page.

This section therefore supports the credibility of the project. It gives enough detail for a reviewer to see how WriteMate was planned, built, and evaluated as a complete assistive writing system. In Section 3.2, this point is specifically applied to non-functional requirements.

3.3 VOICE DATASET

This part of the methodology explains how voice dataset creation was planned and integrated. The project depends on a chain of modules, so the design of one stage must account for the assumptions and limitations of the next.

In this module, 2,600 utterances defines the input condition, 10 speakers affects processing reliability, alphabet coverage supports integration, and 80:20 split influences the final user experience.

Calibration is treated as a repeated process rather than a one-time setting. Pen pressure, motor movement, paper margins, and input confidence need periodic checking to preserve output quality. In Section 3.3, this point is specifically applied to voice dataset creation.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.3, this point is specifically applied to voice dataset creation.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.3, this point is specifically applied to voice dataset creation.

Raspberry Pi 5 is capable, but it cannot be treated like an unlimited server; memory, processing time, and peripheral timing must be managed carefully.

To keep the system responsive, the recognition tasks are focused and the writing commands are structured. The project avoids unnecessary complexity that would increase latency without improving the user's writing experience. In Section 3.3, this point is specifically applied to voice dataset creation.

The relationship between 2,600 utterances and 10 speakers affects the input side, while alphabet coverage and 80:20 split influence how the system behaves after recognition. Both sides must remain synchronized.

This careful sequencing is what makes the prototype suitable for offline operation. In Section 3.3, this point is specifically applied to voice dataset creation.

Section 3.3 also creates traceability between the report objective and the prototype implementation. In the case of voice dataset creation, the discussion connects 2,600 utterances with the early design requirement, 10 speakers with the processing stage, and alphabet coverage with the reliability expected from the final output.

A practical reader should be able to follow how this section affects the system that was built. The explanation of voice dataset creation is therefore linked with the microphone, camera, Raspberry Pi, recognition models, buffer logic, and CNC writer wherever those parts are relevant.

The section further avoids treating accessibility as a vague idea. It translates the need for 80:20 split into concrete design behaviour: the system should respond predictably, preserve user control, and create written output that can be checked on paper.

The project depends on a chain of modules, so the design of one stage must account for the assumptions and limitations of the next.

In this module, noise variation defines the input condition, normalization affects processing reliability, command phrases supports integration, and held-out testing influences the final user experience.

Calibration is treated as a repeated process rather than a one-time setting. Pen pressure, motor movement, paper margins, and input confidence need periodic checking to preserve output quality. In Section 3.3, this point is specifically applied to voice dataset preparation.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.3, this point is specifically applied to voice dataset preparation.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.3, this point is specifically applied to voice dataset preparation.

it cannot be treated like an unlimited server; memory, processing time, and peripheral timing must be managed carefully.

To keep the system responsive, the recognition tasks are focused and the writing commands are structured. The project avoids unnecessary complexity that would increase latency without improving the user's writing experience. The relationship between noise variation and normalization affects the input side, while command phrases and held-out testing influence how the system behaves after recognition. Both sides must remain Another important aspect of Section 3.3 is its connection with error prevention. For voice dataset preparation, the system must reduce mistakes before they reach the writing stage, because errors written on paper are harder to ignore than errors displayed temporarily on a screen.

This is why the report repeatedly connects noise variation and normalization with validation. The device is expected to filter unclear input, stabilize uncertain gestures, and organize recognized text before G-code is generated.

The same reasoning applies to command phrases and held-out testing. They influence how the prototype behaves during repeated use, where consistency matters as much as a single successful demonstration.

3.4 ISL GESTURE DATASET

The development of gesture dataset creation required both software and hardware considerations. Unlike a purely digital application, WriteMate must preserve accuracy through signal capture, model inference, text preparation, and mechanical writing.

The workflow gives special importance to 5,200 samples and 26 classes, because noisy input or unstable recognition would create errors before the CNC mechanism begins writing. The remaining concerns, 8 participants and 30 FPS capture, support consistency during repeated use.

The section shows that the project is not only a machine-learning task. It is an embedded assistive system where data preparation, inference, user interaction, and actuation all contribute to success.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.4, this point is specifically applied to gesture dataset creation.

The methodology therefore supports both reliability and maintainability. If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.4, this point is specifically applied to gesture dataset creation.

The role of 5,200 samples is checked before 26 classes; the results of 8 participants are inspected before 30 FPS capture is executed. This staged flow reduces the chance that an early error will silently become a bad written output.

The methodology therefore supports both development and evaluation. It gives the report a clear basis for explaining where the final results come from.

Section 3.4 is also relevant from a deployment point of view. A college, hospital, or office would judge gesture dataset creation not only by technical novelty but by whether the device can be handled, calibrated, and trusted by ordinary users.

Lighting, background noise, paper alignment, motor tuning, and input confidence all affect the experience even when the core algorithm is correct.

The details involving 5,200 samples, 26 classes, 8 participants, and 30 FPS capture help convert the project from an idea into a usable prototype. They show that the team considered both software behaviour and physical constraints.

In the proposed methodology, gesture preprocessing is treated as a controlled engineering task. The section explains how frames are prepared before classification while keeping the system suitable for offline execution on Raspberry Pi hardware.

The module was designed with a balance between accuracy and simplicity. Features related to background removal, skin-region cues, resizing, and landmark normalization were included because they directly improve the system's ability to operate without manual supervision.

Accuracy, latency, line straightness, angular error, and robustness are all connected to decisions made during implementation.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.4, this point is specifically applied to gesture preprocessing.

In gesture preprocessing, user comfort remains a design consideration even when the section is technical. The system should not ask the user to repeat commands unnecessarily, wait for long processing delays, or correct avoidable writing mistakes.

This is why recognition confidence, stable gesture detection, and clean text buffering are included before motion commands are produced. The final handwriting should represent the user's intention as closely as possible.

The implementation details connected to background removal, skin-region cues, resizing, and landmark normalization are therefore part of the accessibility design, not just internal engineering choices.

Once the module is integrated, it contributes to the larger goal of making writing independent, private, and consistent.

Section 3.4 is also relevant from a deployment point of view. A college, hospital, or office would judge gesture preprocessing not only by technical novelty but by whether the device can be handled, calibrated, and trusted by ordinary users.

The report therefore explains how frames are prepared before classification with attention to operating conditions. Lighting, background noise, paper alignment, motor tuning, and input confidence all affect the experience even when the core algorithm is correct.

The details involving background removal, skin-region cues, resizing, and landmark normalization help convert the project from an idea into a usable prototype. They show that the team considered both software behaviour and physical constraints.

3.5 SPEECH MODULE

In the proposed methodology, the speech recognition module is treated as a controlled engineering task. The section explains how to describe the offline conversion from spoken input to text while keeping the system suitable for offline execution on Raspberry Pi hardware.

The module was designed with a balance between accuracy and simplicity. Features related to Vosk model, filtered audio, recognized tokens, and text buffer were included because they directly improve the system's ability to operate without manual supervision.

Accuracy, latency, line straightness, angular error, and robustness are all connected to decisions made during implementation. In Section 3.5, this point is specifically applied to the speech recognition module.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.5, this point is specifically applied to the speech recognition module.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.5, this point is specifically applied to the speech recognition module.

In the speech recognition module, user comfort remains a design consideration even when the section is technical. The system should not ask the user to repeat commands unnecessarily, wait for long processing delays, or correct avoidable writing mistakes.

This is why recognition confidence, stable gesture detection, and clean text buffering are included before motion commands are produced. The final handwriting should represent the user's intention as closely as possible. In Section 3.5, this point is specifically applied to the speech recognition module.

Once the module is integrated, it contributes to the larger goal of making writing independent, private, and consistent. In Section 3.5, this point is specifically applied to the speech recognition module.

From a documentation perspective, Section 3.5 helps show that the project decisions are not arbitrary. The treatment of speech recognition module follows from the problem statement and leads naturally into the design, testing, or conclusion that appears later in the report.

The connection between Vosk model and text buffer is especially useful because it shows the movement from input requirement to output requirement. WriteMate begins with user expression and ends with handwriting, so both ends of the pipeline must be discussed.

By including filtered audio and recognized tokens, the section also records the intermediate engineering decisions. These decisions make the prototype more reliable than a simple direct connection between a recognizer and a plotter.

This section focuses on explaining how invalid or noisy recognition results are handled, and it explains the practical decisions taken while building the WriteMate pipeline.

The implementation pays attention to allowed vocabulary, confidence checks, repeat commands, and error reduction. These points influence the module's behaviour under real conditions rather than only in ideal demonstrations.

because an error can be traced to the speech module, gesture module, buffer, G-code generator, or motion system. In Section 3.5, this point is specifically applied to speech module validation.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.5, this point is specifically applied to speech module validation.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.5, this point is specifically applied to speech module validation.

If recognition failed, the team could inspect the input and model stages; if handwriting failed, the G-code or mechanical calibration could be examined separately.

This separation is important because WriteMate combines different engineering domains. Audio processing, computer vision, embedded programming, and CNC control each have their own failure modes. In Section 3.5, this point is specifically applied to speech module validation.

The points allowed vocabulary, confidence checks, repeat commands, and error reduction were therefore translated into module-level responsibilities. Each module produces an output that can be inspected before it is passed forward.

Such modular development also supports future improvement. A better speech model or gesture classifier can be added without changing the entire writing mechanism. In Section 3.5, this point is specifically applied to speech module validation.

In the case of speech module validation, the discussion connects allowed vocabulary with the early design requirement, confidence checks with the processing stage, and repeat commands with the reliability expected from the final output.

A practical reader should be able to follow how this section affects the system that was built. The explanation of speech module validation is therefore linked with the microphone, camera, Raspberry Pi, recognition models, buffer logic, and CNC writer wherever those parts are relevant.

The section further avoids treating accessibility as a vague idea. It translates the need for error reduction into concrete design behaviour: the system should respond predictably, preserve user control, and create written output that can be checked on paper.

3.6 GESTURE MODULE

Unlike a purely digital application, WriteMate must preserve accuracy through signal capture, model inference, text preparation, and mechanical writing.

The workflow gives special importance to camera capture and MediaPipe hands, because noisy input or unstable recognition would create errors before the CNC mechanism begins writing. The remaining concerns, YOLOv8 classifier and gesture label mapping, support consistency during repeated use.

The section shows that the project is not only a machine-learning task. It is an embedded assistive system where data preparation, inference, user interaction, and actuation all contribute to success. In Section 3.6, this point is specifically applied to gesture recognition module.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line The design of gesture recognition module was also shaped by the need for simple troubleshooting. When a prototype includes motors, sensors, cameras, and software models, problems can appear in many places.

By separating capture, recognition, buffering, G-code generation, and execution, the team can test each stage independently. This makes the project easier to demonstrate, maintain, and extend. In Section 3.6, this point is specifically applied to gesture recognition module.

The role of camera capture is checked before MediaPipe hands; the results of YOLOv8 classifier are inspected before gesture label mapping is executed. This staged flow reduces the chance that an early error will silently become a bad written output.

The details involving camera capture, MediaPipe hands, YOLOv8 classifier, and gesture label mapping help convert the project from an idea into a usable prototype. They show that the team considered both software behaviour and physical constraints.

In the proposed methodology, gesture ambiguity handling is treated as a controlled engineering task. The section explains how explaining how similar signs are stabilized while keeping the system suitable for offline execution on Raspberry Pi hardware.

The module was designed with a balance between accuracy and simplicity. Features related to frame sequence, confidence threshold, temporal smoothing, and no-gesture state were included because they directly improve the system's ability to operate without manual supervision.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.6, this point is specifically applied to gesture ambiguity handling.

The methodology therefore supports both reliability and maintainability. If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.6, this point is specifically applied to gesture ambiguity handling.

In gesture ambiguity handling, user comfort remains a design consideration even when the section is technical. The system should not ask the user to repeat commands unnecessarily, wait for long processing delays, or correct avoidable writing mistakes.

This is why recognition confidence, stable gesture detection, and clean text buffering are included before motion commands are produced. The final handwriting should represent the user's intention as closely as possible. In Section 3.6, this point is specifically applied to gesture ambiguity handling.

The implementation details connected to frame sequence, confidence threshold, temporal smoothing, and no-gesture state are therefore part of the accessibility design, not just internal engineering choices.

Once the module is integrated, it contributes to the larger goal of making writing independent, private, and consistent. In Section 3.6, this point is specifically applied to gesture ambiguity handling.

For this reason, the report gives attention to frame sequence, confidence threshold, temporal smoothing, and no-gesture state rather than presenting only final results. Each point contributes to the quality of the handwritten page.

This section therefore supports the credibility of the project. It gives enough detail for a reviewer to see how WriteMate was planned, built, and evaluated as a complete assistive writing system. In Section 3.6, this point is specifically applied to gesture ambiguity handling.

3.7 TEXT BUFFER MANAGER

This section documents the construction of text buffer management in enough detail to show how the prototype can be reproduced or improved. The focus is practical: data, processing, calibration, and integration are described as working steps.

Practical constraints shaped the implementation. The Raspberry Pi must process input quickly, the recognition modules must avoid unnecessary computation, and the CNC controller must receive commands that are already organized for writing. In Section 3.7, this point is specifically applied to text buffer management.

By documenting this module carefully, the report makes the prototype understandable as a system rather than as a collection of unrelated components. In Section 3.7, this point is specifically applied to text buffer management.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.7, this point is specifically applied to text buffer management.

For example, the gesture classifier can be retrained with more samples, the speech vocabulary can be expanded, or the stroke library can be modified for new scripts. These changes are possible because the pipeline is not tightly locked to one model. In Section 3.7, this point is specifically applied to text buffer management.

The design points spacing, line wrapping, correction commands, and punctuation handling therefore act as extension points as well as current implementation details.

This makes the project report useful not only as a record of completed work, but also as documentation for continuing development. In Section 3.7, this point is specifically applied to text buffer management.

This section therefore supports the credibility of the project. It gives enough detail for a reviewer to see how WriteMate was planned, built, and evaluated as a complete assistive writing system. In Section 3.7, this point is specifically applied to text buffer management.

3.8 CNC ASSEMBLY

CNC mechanism assembly is handled as an implementation activity with measurable outputs. The design is not limited to diagrams; each module has to accept input, process it correctly, and pass a clean result to the next stage.

The design choices associated with this section include X-Y axis, stepper motors, belt alignment, and paper positioning. Each choice was made to reduce ambiguity before the final handwriting stage, where even a small recognition error becomes visible on paper.

A modular structure was maintained throughout development. This allows the speech model, gesture classifier, or stroke library to be upgraded later without redesigning the entire system.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.8, this point is specifically applied to cNC mechanism assembly.

For cNC mechanism assembly, the practical challenge is not only making the module work once, but making it behave predictably across repeated attempts. Assistive devices need repeatability because users may rely on them during important tasks.

The methodology therefore includes preprocessing and validation rather than sending raw recognition output directly to the CNC unit. This protects the writing stage from avoidable input errors.

The design attention to X-Y axis, stepper motors, belt alignment, and paper positioning helps keep the workflow stable. These details may appear small individually, but together they determine whether the written page is usable.

The same reasoning applies to calibration. Mechanical settings are tested and adjusted because small movement errors become visible in handwriting.

Another important aspect of Section 3.8 is its connection with error prevention. For cNC mechanism assembly, the system must reduce mistakes before they reach the writing stage, because errors written on paper are harder to ignore than errors displayed temporarily on a screen.

This part of the methodology explains how the pen-lift mechanism was planned and integrated. The project depends on a chain of modules, so the design of one stage must account for the assumptions and limitations of the next.

In this module, a servo motor defines the input condition, pen pressure affects processing reliability, pen-up movement supports integration, and clean strokes influences the final user experience.

Calibration is treated as a repeated process rather than a one-time setting. Pen pressure, motor movement, paper margins, and input confidence need periodic checking to preserve output quality. In Section 3.8, this point is specifically applied to pen-lift mechanism.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.8, this point is specifically applied to the pen-lift mechanism.

To keep the system responsive, the recognition tasks are focused and the writing commands are structured. The project avoids unnecessary complexity that would increase latency without improving the user's writing experience. In Section 3.8, this point is specifically applied to pen-lift mechanism.

The relationship between servo motor and pen pressure affects the input side, while pen-up movement and clean strokes influence how the system behaves after recognition. Both sides must remain synchronized.

This careful sequencing is what makes the prototype suitable for offline operation. In Section 3.8, this point is specifically applied to pen-lift mechanism.

Section 3.8 also creates traceability between the report objective and the prototype implementation. In the case of pen-lift mechanism, the discussion connects servo motor with the early design requirement, pen pressure with the processing stage, and pen-up movement with the reliability expected from the final output.

3.9 MOTOR CALIBRATION

This section documents the construction of motor calibration in enough detail to show how the prototype can be reproduced or improved. The focus is practical: data, processing, calibration, and integration are described as working steps.

Practical constraints shaped the implementation. The Raspberry Pi must process input quickly, the recognition modules must avoid unnecessary computation, and the CNC controller must receive commands that are already organized for writing. In Section 3.9, this point is specifically applied to motor calibration.

By documenting this module carefully, the report makes the prototype understandable as a system rather than as a collection of unrelated components. In Section 3.9, this point is specifically applied to motor calibration.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.9, this point is specifically applied to motor calibration.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.9, this point is specifically applied to motor calibration.

A clear module boundary around motor calibration makes later replacement and experimentation easier.

For example, the gesture classifier can be retrained with more samples, the speech vocabulary can be expanded, or the stroke library can be modified for new scripts. These changes are possible because the pipeline is not tightly locked to one model. In Section 3.9, this point is specifically applied to motor calibration.

The design points microstepping, acceleration, home position, and test strokes therefore act as extension points as well as current implementation details.

By including acceleration and home position, the section also records the intermediate engineering decisions.

3.10 G-CODE GENERATION

The development of g-code generation required both software and hardware considerations. Unlike a purely digital application, WriteMate must preserve accuracy through signal capture, model inference, text preparation, and mechanical writing.

The workflow gives special importance to character strokes and G0 commands, because noisy input or unstable recognition would create errors before the CNC mechanism begins writing. The remaining concerns, G1 commands and pen control, support consistency during repeated use.

The section shows that the project is not only a machine-learning task. It is an embedded assistive system where data preparation, inference, user interaction, and actuation all contribute to success. In Section 3.10, this point is specifically applied to g-code generation.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.10, this point is specifically applied to g-code generation.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.10, this point is specifically applied to g-code generation.

The design of g-code generation was also shaped by the need for simple troubleshooting. When a prototype includes motors, sensors, cameras, and software models, problems can appear in many places.

By separating capture, recognition, buffering, G-code generation, and execution, the team can test each stage independently. This makes the project easier to demonstrate, maintain, and extend. In Section 3.10, this point is specifically applied to g-code generation.

The role of character strokes is checked before G0 commands; the results of G1 commands are inspected before pen control is executed. This staged flow reduces the chance that an early error will silently become a bad written output.

The development of stroke path design required both software and hardware considerations. Unlike a purely digital application, WriteMate must preserve accuracy through signal capture, model inference, text preparation, and mechanical writing.

The workflow gives special importance to line segments and curves, because noisy input or unstable recognition would create errors before the CNC mechanism begins writing. The remaining concerns, stroke order and page margins, support consistency during repeated It is an embedded assistive system where data preparation, inference, user interaction, and actuation all contribute to success. In Section 3.10, this point is specifically applied to stroke path design.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.10, this point is specifically applied to stroke path design.

The methodology therefore supports both reliability and maintainability. If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.10, this point is specifically applied to stroke path design.

By separating capture, recognition, buffering, G-code generation, and execution, the team can test each stage independently. This makes the project easier to demonstrate, maintain, and extend. In Section 3.10, this point is specifically applied to stroke path design.

The role of line segments is checked before curves; the results of stroke order are inspected before page margins is executed. This staged flow reduces the chance that an early error will silently become a bad written output.

The methodology therefore supports both development and evaluation. It gives the report a clear basis for explaining where the final results come from. In Section 3.10, this point is specifically applied to stroke path design.

The discussion in Section 3.10 can also be read as a design checkpoint. If stroke path design is weak, the rest of the system becomes less dependable, because later modules assume that earlier decisions have produced clean and meaningful information.

For this reason, the report gives attention to line segments, curves, stroke order, and page margins rather than presenting only final results. Each point contributes to the quality of the handwritten page.

This section therefore supports the credibility of the project. It gives enough detail for a reviewer to see how WriteMate was planned, built, and evaluated as a complete assistive writing system. In Section 3.10, this point is specifically applied to stroke path design.

3.11 SYSTEM ARCHITECTURE

In the proposed methodology, the input acquisition layer is treated as a controlled engineering task. The section explains how to explain the first layer of the architecture while keeping the system suitable for offline execution on Raspberry Pi hardware.

The module was designed with a balance between accuracy and simplicity. Features related to microphone, web camera, preprocessing, and feature-ready signals were included because they directly improve the system's ability to operate without manual supervision.

Accuracy, latency, line straightness, angular error, and robustness are all connected to decisions made during implementation. In Section 3.11, this point is specifically applied to input acquisition layer.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.11, this point is specifically applied to input acquisition layer.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.11, this point is specifically applied to input acquisition layer.

In input acquisition layer, user comfort remains a design consideration even when the section is technical. The system should not ask the user to repeat commands unnecessarily, wait for long processing delays, or correct avoidable writing mistakes.

This is why recognition confidence, stable gesture detection, and clean text buffering are included before motion commands are produced. The final handwriting should represent the user's intention as closely as possible. In Section 3.11, this point is specifically applied to input acquisition layer.

The implementation details connected to microphone, web camera, preprocessing, and feature-ready signals are therefore part of the accessibility design, not just internal engineering choices.

Once the module is integrated, it contributes to the larger goal of making writing independent, private, and consistent. In Section 3.11, this point is specifically applied to input acquisition layer.

The discussion in Section 3.11 can also be read as a design checkpoint. If input acquisition layer is weak, the rest of the system becomes less dependable, because later modules assume that earlier decisions have produced clean and meaningful information.

For this reason, the report gives attention to microphone, web camera, preprocessing, and feature-ready signals rather than presenting only final results. Each point contributes to the quality of the handwritten page.

This section therefore supports the credibility of the project. It gives enough detail for a reviewer to see how WriteMate was planned, built, and evaluated as a complete assistive writing system. In Section 3.11, this point is specifically applied to the input acquisition layer.

This section focuses on explaining the intelligent decision layer, and it explains the practical decisions taken while building the WriteMate pipeline.

The implementation pays attention to ASR, gesture classifier, buffer manager, and G-code conversion. These points influence the module's behaviour under real conditions rather than only in ideal demonstrations.

This separation makes testing clearer, because an error can be traced to the speech module, gesture module, buffer, G-code generator, or motion system. In Section 3.11, this point is specifically applied to processing layer.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.11, this point is specifically applied to processing layer.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.11, this point is specifically applied to processing layer.

The implementation of processing layer was planned so that errors could be traced clearly. If recognition failed, the team could inspect the input and model stages; if handwriting failed, the G-code or mechanical calibration could be examined separately.

This separation is important because WriteMate combines different engineering domains. Audio processing, computer vision, embedded programming, and CNC control each have

their own failure modes. In Section 3.11, this point is specifically applied to processing layer.

The points ASR, gesture classifier, buffer manager, and G-code conversion were therefore translated into module-level responsibilities. Each module produces an output that can be inspected before it is passed forward.

Such modular development also supports future improvement. A better speech model or gesture classifier can be added without changing the entire writing mechanism. In Section 3.11, this point is specifically applied to processing layer.

If processing layer is weak, the rest of the system becomes less dependable, because later modules assume that earlier decisions have produced clean and meaningful information.

The section explains how explaining the conversion of commands into handwriting while keeping the system suitable for offline execution on Raspberry Pi hardware.

The module was designed with a balance between accuracy and simplicity. Features related to interpreter, motor pulses, servo control, and instruction queue were included because they directly improve the system's ability to operate without manual supervision.

These methodological choices support the evaluation described in the next chapter. Accuracy, latency, line straightness, angular error, and robustness are all connected to decisions made during implementation. In Section 3.11, this point is specifically applied to output execution layer.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.11, this point is specifically applied to output execution layer.

If future work adds regional language scripts, personalized handwriting, or improved gesture models, the same staged pipeline can be extended without changing the purpose of the device. In Section 3.11, this point is specifically applied to output execution layer.

In output execution layer, user comfort remains a design consideration even when the section is technical. The system should not ask the user to repeat commands unnecessarily, wait for long processing delays, or correct avoidable writing mistakes.

This is why recognition confidence, stable gesture detection, and clean text buffering are included before motion commands are produced. The final handwriting should represent the

user's intention as closely as possible. In Section 3.11, this point is specifically applied to output execution layer.

The implementation details connected to interpreter, motor pulses, servo control, and instruction queue are therefore part of the accessibility design, not just internal engineering choices.

Once the module is integrated, it contributes to the larger goal of making writing independent, private, and consistent. In Section 3.11, this point is specifically applied to output execution layer.

Section 3.11 also creates traceability between the report objective and the prototype implementation. In the case of output execution layer, the discussion connects interpreter with the early design requirement, motor pulses with the processing stage, and servo control with the reliability expected from the final output.

A practical reader should be able to follow how this section affects the system that was built. The explanation of output execution layer is therefore linked with the microphone, camera, Raspberry Pi, recognition models, buffer logic, and CNC writer wherever those parts are relevant.

The section further avoids treating accessibility as a vague idea. It translates the need for instruction queue into concrete design behaviour: the system should respond predictably, preserve user control, and create written output that can be checked on paper.

The development of user interface layer required both software and hardware considerations. Unlike a purely digital application, WriteMate must preserve accuracy through signal capture, model inference, text preparation, and mechanical writing.

The workflow gives special importance to mode selection and feedback, because noisy input or unstable recognition would create errors before the CNC mechanism begins writing. The remaining concerns, paper configuration and calibration panel, support consistency during repeated use.

The section shows that the project is not only a machine-learning task. It is an embedded assistive system where data preparation, inference, user interaction, and actuation all contribute to success. In Section 3.11, this point is specifically applied to user interface layer.

The design of user interface layer was also shaped by the need for simple troubleshooting. When a prototype includes motors, sensors, cameras, and software models, problems can appear in many places.

By separating capture, recognition, buffering, G-code generation, and execution, the team can test each stage independently. This makes the project easier to demonstrate, maintain, and extend. In Section 3.11, this point is specifically applied to user interface layer.

This staged flow reduces the chance that an early error will silently become a bad written output.

It gives the report a clear basis for explaining where the final results come from. In Section 3.11, this point is specifically applied to user interface layer.

From a documentation perspective, Section 3.11 helps show that the project decisions are not arbitrary. The treatment of user interface layer follows from the problem statement and leads naturally into the design, testing, or conclusion that appears later in the report.

WriteMate begins with user expression and ends with handwriting, so both ends of the pipeline must be discussed.

By including feedback and paper configuration, the section also records the intermediate engineering decisions. These decisions make the prototype more reliable than a simple direct connection between a recognizer and a plotter.

3.12 SYSTEM WORKFLOW

End-to-end workflow is handled as an implementation activity with measurable outputs. The design is not limited to diagrams; each module has to accept input, process it correctly, and pass a clean result to the next stage.

The design choices associated with this section include mode selection, recognition, validation, and handwriting execution. Each choice was made to reduce ambiguity before the final handwriting stage, where even a small recognition error becomes visible on paper.

The integration sequence is especially important. Recognized text is not sent directly to the motors; it first passes through validation and buffering so that spacing, correction, and line flow can be handled before G-code is produced. In Section 3.12, this point is specifically applied to end-to-end workflow.

The same reasoning applies to calibration. Mechanical settings are tested and adjusted because small movement errors become visible in handwriting. In Section 3.12, this point is specifically applied to end-to-end workflow.

CHAPTER 4
RESULTS AND DISCUSSION

4.1 TESTING STRATEGY

Testing strategy is discussed in this chapter as evidence of practical performance. The purpose of testing is to determine whether the system can work reliably enough for assistive writing, not simply whether individual modules run.

The reported values related to model accuracy, latency, writing precision, and robustness show that the prototype meets the main performance expectations. The results are strongest when the input is clear and the writing mechanism has been properly calibrated.

The comparison with related systems is important because it shows that WriteMate is not only accurate but also different in output type. Producing physical handwriting offline is the key distinction.

The test results should be interpreted in relation to the project's scope. The prototype focuses on English alphabet recognition, command phrases, and CNC handwriting, so the reported numbers represent this controlled but relevant use case.

The evaluation also suggests how future deployments should be tested: with more users, varied lighting, different noise levels, longer writing tasks, and repeated calibration cycles across multiple days.

In evaluating testing strategy, the report considers the user's experience of delay and correction. If the system responds slowly, the writing process becomes frustrating; if it writes incorrect content, the user must spend additional effort correcting it.

The sub-150 ms target is therefore not arbitrary. It reflects the need for interaction that feels responsive enough for continuous use.

The relationship between model accuracy and latency reveals the recognition side of the system, while writing precision and robustness describe the quality and stability of output.

Together, these measures indicate that WriteMate can serve as a practical prototype rather than only a laboratory exercise.

The discussion in Section 4.1 can also be read as a design checkpoint. If testing strategy is weak, the rest of the system becomes less dependable, because later modules assume that earlier decisions have produced clean and meaningful information.

For this reason, the report gives attention to model accuracy, latency, writing precision, and robustness rather than presenting only final results. Each point contributes to the quality of the handwritten page.

This section therefore supports the credibility of the project. It gives enough detail for a reviewer to see how WriteMate was planned, built, and evaluated as a complete assistive writing system. In Section 4.1, this point is specifically applied to testing strategy.

4.2 MODEL ACCURACY

In this chapter, model accuracy testing is examined as part of the complete system behaviour. Recognition, buffering, and CNC execution are all considered because the final user sees only the written result.

The result analysis therefore combines numerical and practical observations. Metrics such as F1 score, word error rate, and line deviation are interpreted alongside the usability requirement of producing clean handwriting.

These findings make the prototype suitable as a foundation for future development. It already demonstrates feasibility while leaving room for speed, language, and personalization improvements.

The test results should be interpreted in relation to the project's scope. The prototype focuses on English alphabet recognition, command phrases, and CNC handwriting, so the reported numbers represent this controlled but relevant use case. In Section 4.2, this point is specifically applied to model accuracy testing.

The evaluation also suggests how future deployments should be tested: with more users, varied lighting, different noise levels, longer writing tasks, and repeated calibration cycles across multiple days. In Section 4.2, this point is specifically applied to model accuracy testing.

For model accuracy testing, robustness is as important as peak performance. A device used in an examination or documentation setting must continue operating after the first few minutes of demonstration.

The long-duration tests show that the prototype can maintain stroke consistency across extended sessions. This does not eliminate the need for field trials, but it strengthens confidence in the mechanical design.

The role of per-class accuracy and macro F1 is visible in immediate response, while WER and confusion patterns become more important during longer writing tasks.

Such evaluation is necessary because assistive technology must be dependable under repeated use.

By including macro F1 and WER, the section also records the intermediate engineering decisions. These decisions make the prototype more reliable than a simple direct connection between a recognizer and a plotter.

4.3 LETTER ACCURACY

Letter recognition results is discussed in this chapter as evidence of practical performance. The purpose of testing is to determine whether the system can work reliably enough for assistive writing, not simply whether individual modules run.

The reported values related to C and D accuracy, Z difficulty, visual similarity, and alphabet coverage show that the prototype meets the main performance expectations. The results are strongest when the input is clear and the writing mechanism has been properly calibrated.

The comparison with related systems is important because it shows that WriteMate is not only accurate but also different in output type. Producing physical handwriting offline is the key distinction. In Section 4.3, this point is specifically applied to letter recognition results.

The test results should be interpreted in relation to the project's scope. The prototype focuses on English alphabet recognition, command phrases, and CNC handwriting, so the reported numbers represent this controlled but relevant use case. In Section 4.3, this point is specifically applied to letter recognition results.

The evaluation also suggests how future deployments should be tested: with more users, varied lighting, different noise levels, longer writing tasks, and repeated calibration cycles across multiple days. In Section 4.3, this point is specifically applied to letter recognition results.

In evaluating letter recognition results, the report considers the user's experience of delay and correction. If the system responds slowly, the writing process becomes frustrating; if it writes incorrect content, the user must spend additional effort correcting it.

It reflects the need for interaction that feels responsive enough for continuous use. In Section 4.3, this point is specifically applied to letter recognition results.

Together, these measures indicate that WriteMate can serve as a practical prototype rather than only a laboratory exercise. In Section 4.3, this point is specifically applied to letter recognition results.

By including Z difficulty and visual similarity, the section also records the intermediate engineering decisions. These decisions make the prototype more reliable than a simple direct connection between a recognizer and a plotter.

Table 2

Letter	Acc. (%)	Letter	Acc. (%)	Letter	Acc. (%)
A	94	J	93	S	94
B	94	K	93	T	93
C	95	L	93	U	93
D	95	M	93	V	90
E	90	N	93	W	94
F	92	O	92	X	91
G	93	P	94	Y	92
H	93	Q	90	Z	89
I	94	R	93		

4.4 LATENCY

Inference latency is discussed in this chapter as evidence of practical performance. The purpose of testing is to determine whether the system can work reliably enough for assistive writing, not simply whether individual modules run.

The reported values related to 121 ms speech latency, 138 ms gesture latency, sub-150 ms target, and real-time use show that the prototype meets the main performance expectations. The results are strongest when the input is clear and the writing mechanism has been properly calibrated.

The comparison with related systems is important because it shows that WriteMate is not only accurate but also different in output type. Producing physical handwriting offline is the key distinction. In Section 4.4, this point is specifically applied to inference latency.

The test results should be interpreted in relation to the project's scope. The prototype focuses on English alphabet recognition, command phrases, and CNC handwriting, so the reported

numbers represent this controlled but relevant use case. In Section 4.4, this point is specifically applied to inference latency.

The evaluation also suggests how future deployments should be tested: with more users, varied lighting, different noise levels, longer writing tasks, and repeated calibration cycles across multiple days. In Section 4.4, this point is specifically applied to inference latency.

In evaluating inference latency, the report considers the user's experience of delay and correction. If the system responds slowly, the writing process becomes frustrating; if it writes incorrect content, the user must spend additional effort correcting it.

It reflects the need for interaction that feels responsive enough for continuous use. In Section 4.4, this point is specifically applied to inference latency.

The relationship between 121 ms speech latency and 138 ms gesture latency reveals the recognition side of the system, while sub-150 ms target and real-time use describe the quality and stability of output.

Together, these measures indicate that WriteMate can serve as a practical prototype rather than only a laboratory exercise. In Section 4.4, this point is specifically applied to inference latency.

In the case of inference latency, the discussion connects 121 ms speech latency with the early design requirement, 138 ms gesture latency with the processing stage, and sub-150 ms target with the reliability expected from the final output.

A practical reader should be able to follow how this section affects the system that was built. The explanation of inference latency is therefore linked with the microphone, camera, Raspberry Pi, recognition models, buffer logic, and CNC writer wherever those parts are relevant.

It translates the need for real-time use into concrete design behaviour: the system should respond predictably, preserve user control, and create written output that can be checked on paper.

4.5 CNC PRECISION

Writing precision is discussed in this chapter as evidence of practical performance. The purpose of testing is to determine whether the system can work reliably enough for assistive writing, not simply whether individual modules run.

The reported values related to 0.4 mm deviation, 2 degree angular error, stroke overlap, and legibility show that the prototype meets the main performance expectations. The results are strongest when the input is clear and the writing mechanism has been properly calibrated.

The comparison with related systems is important because it shows that WriteMate is not only accurate but also different in output type. Producing physical handwriting offline is the key distinction. In Section 4.5, this point is specifically applied to writing precision.

The test results should be interpreted in relation to the project's scope. The prototype focuses on English alphabet recognition, command phrases, and CNC handwriting, so the reported numbers represent this controlled but relevant use case. In Section 4.5, this point is specifically applied to writing precision.

The evaluation also suggests how future deployments should be tested: with more users, varied lighting, different noise levels, longer writing tasks, and repeated calibration cycles across multiple days. In Section 4.5, this point is specifically applied to writing precision.

If the system responds slowly, the writing process becomes frustrating; if it writes incorrect content, the user must spend additional effort correcting it.

The relationship between 0.4 mm deviation and 2 degree angular error reveals the recognition side of the system, while stroke overlap and legibility describe the quality and stability of output.

Together, these measures indicate that WriteMate can serve as a practical prototype rather than only a laboratory exercise. In Section 4.5, this point is specifically applied to writing precision.

In the case of writing precision, the discussion connects 0.4 mm deviation with the early design requirement, 2 degree angular error with the processing stage, and stroke overlap with the reliability expected from the final output.

A practical reader should be able to follow how this section affects the system that was built. The explanation of writing precision is therefore linked with the microphone, camera, Raspberry Pi, recognition models, buffer logic, and CNC writer wherever those parts are relevant.

The section further avoids treating accessibility as a vague idea. It translates the need for legibility into concrete design behaviour: the system should respond predictably, preserve user control, and create written output that can be checked on paper.

4.6 ROBUSTNESS

Long-duration testing is discussed in this chapter as evidence of practical performance. The purpose of testing is to determine whether the system can work reliably enough for assistive writing, not simply whether individual modules run.

The reported values related to 2-3 hour sessions, no overheating, no motion drift, and multi-page writing show that the prototype meets the main performance expectations. The results are strongest when the input is clear and the writing mechanism has been properly calibrated.

The evaluation also suggests how future deployments should be tested: with more users, varied lighting, different noise levels, longer writing tasks, and repeated calibration cycles across multiple days. In Section 4.6, this point is specifically applied to long-duration testing.

In evaluating long-duration testing, the report considers the user's experience of delay and correction. If the system responds slowly, the writing process becomes frustrating; if it writes incorrect content, the user must spend additional effort correcting it.

The sub-150 ms target is therefore not arbitrary. It reflects the need for interaction that feels responsive enough for continuous use. In Section 4.6, this point is specifically applied to long-duration testing.

The discussion in Section 4.6 can also be read as a design checkpoint. If long-duration testing is weak, the rest of the system becomes less dependable, because later modules assume that earlier decisions have produced clean and meaningful information.

For this reason, the report gives attention to 2-3 hour sessions, no overheating, no motion drift, and multi-page writing rather than presenting only final results. Each point contributes to the quality of the handwritten page.

This section therefore supports the credibility of the project. It gives enough detail for a reviewer to see how WriteMate was planned, built, and evaluated as a complete assistive writing system. In Section 4.6, this point is specifically applied to long-duration testing.

4.7 COMPARISON

This section interprets comparison with related systems with respect to real use. A result is meaningful only when it supports independent writing, acceptable response time, and readable output on paper.

In the test process, dual modality and offline processing indicate recognition performance, while physical output and competitive accuracy reveal how well the physical output stage behaves.The discussion of comparison with related systems also identifies where errors are most likely to occur. Similar hand shapes, unclear speech, poor lighting, and mechanical misalignment can each affect the final result.

Because the system is modular, these errors can be addressed separately. Dataset expansion may improve recognition, while motor tuning and better fixtures may improve writing quality.

The evidence related to dual modality, offline processing, physical output, and competitive accuracy helps prioritize these improvements. Future work should focus first on the issues that most directly affect user trust.

This interpretation keeps the evaluation constructive rather than merely descriptive.

The connection between dual modality and competitive accuracy is especially useful because it shows the movement from input requirement to output requirement. WriteMate begins with user expression and ends with handwriting, so both ends of the pipeline must be discussed.

By including offline processing and physical output, the section also records the intermediate engineering decisions. These decisions make the prototype more reliable than a simple direct connection between a recognizer and a plotter.

Table 3

System	Input	Offline	Output	Accuracy
Desai & Rungta [15]	Voice only	No	Physical	N/R
Katoch et al. [1]	ISL only	Yes	Digital	88.9%
DeepSign [2]	Gesture only	Yes	Digital	91.2%
WriteMate	Voice + ISL	Yes	Physical	92.7%

4.8 DISCUSSION

The results for practical discussion help judge whether the selected methods are suitable for the intended environment. Examinations and documentation tasks require both correctness and consistency.

The values obtained from the prototype are consistent with the aim of a low-cost embedded assistive system. They show that careful integration can produce useful performance without relying on expensive dedicated hardware.

Overall, the result chapter shows that WriteMate's design choices are aligned with its accessibility objective: dependable offline recognition followed by consistent physical writing.

The test results should be interpreted in relation to the project's scope. The prototype focuses on English alphabet recognition, command phrases, and CNC handwriting, so the reported numbers represent this controlled but relevant use case. In Section 4.8, this point is specifically applied to practical discussion , different noise levels, longer writing tasks, and repeated calibration cycles across multiple days. In Section 4.8, this point is specifically applied to practical discussion.

The results connected with practical discussion also help define future work. They show which parts of the prototype are already strong and which parts require refinement before broader deployment.

The outcome of testing examination centres, hospitals, administrative offices, and inclusive education therefore becomes a roadmap. It suggests model improvements, interface improvements, and mechanical improvements at the same time.

The report therefore explains connecting test results to real deployment with attention to operating conditions. Lighting, background noise, paper alignment, motor tuning, and input confidence all affect the experience even when the core algorithm is correct.

The details involving examination centres, hospitals, administrative offices, and inclusive education help convert the project from an idea into a usable prototype. They show that the team considered both software behaviour and physical constraints.

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

Conclusion is discussed here as a summary of what WriteMate achieves and what remains open for improvement. The system demonstrates that accessible input can be converted into physical handwriting with affordable hardware.

The project demonstrates progress in multimodal recognition and embedded deployment, while CNC handwriting and user independence point toward the next stage of refinement.

A major future opportunity is regional language support. For Indian deployment, the ability to write Hindi and other scripts would increase the practical value of the system significantly.

The completed work also provides a useful academic foundation. It combines machine learning, computer vision, embedded systems, and mechatronics in one socially purposeful application.

Future development should preserve the project's central values: offline operation, affordability, user control, and output that can be trusted in formal settings.

The future direction for conclusion should preserve the offline and user-controlled nature of the system. Improvements should not make the device dependent on expensive infrastructure or continuous internet access.

From a documentation perspective, Section 5.1 helps show that the project decisions are not arbitrary. The treatment of conclusion follows from the problem statement and leads naturally into the design, testing, or conclusion that appears later in the report.

The connection between multimodal recognition and user independence is especially useful because it shows the movement from input requirement to output requirement. WriteMate begins with user expression and ends with handwriting, so both ends of the pipeline must be discussed.

By including embedded deployment and CNC handwriting, the section also records the intermediate engineering decisions. These decisions make the prototype more reliable than a simple direct connection between a recognizer and a plotter.

This final chapter uses outcome summary to connect the project results with future development. The prototype is successful as a proof of concept, while also showing clear directions for broader deployment.

From the completed work, it is clear that 92.7 percent accuracy cannot be considered separately from 0.924 macro F1. The final impact depends on the whole chain, including sub-150 ms latency and consistent handwriting.

Hardware refinement can also improve adoption. A compact enclosure, faster motor control, safer pen mounting, and easier calibration would make the device more suitable for institutional use.

The completed work also provides a useful academic foundation. It combines machine learning, computer vision, embedded systems, and mechatronics in one socially purposeful application. In Section 5.1, this point is specifically applied to outcome summary.

Future development should preserve the project's central values: offline operation, affordability, user control, and output that can be trusted in formal settings. In Section 5.1, this point is specifically applied to outcome summary.

In relation to outcome summary, the final report also recognizes the need for user testing. Laboratory results are valuable, but the strongest evaluation would involve users who actually face writing barriers.

Feedback from such users could improve the interface, input timing, correction options, page layout, and physical positioning of the device.

The achieved work around 92.7 percent accuracy and 0.924 macro F1 provides a base for this testing, while improvements in sub-150 ms latency and consistent handwriting would make the prototype easier to use in varied settings.

This user-centred future work is essential for turning the project from a successful academic prototype into a reliable assistive product.

Another important aspect of Section 5.1 is its connection with error prevention. For outcome summary, the system must reduce mistakes before they reach the writing stage, because errors written on paper are harder to ignore than errors displayed temporarily on a screen.

5.2 FUTURE SCOPE

The conclusion on language and personalization scope brings together the technical and social outcomes of the project. This section focuses on identifying improvements for broader adoption and explains how the completed prototype addresses the original problem.

The important outcomes include regional scripts, personal writing style, speaker adaptation, and gesture personalization. These outcomes show that the design is technically feasible and socially relevant.

The system works as an integrated prototype, but future versions should be tested with more users and longer real-world writing tasks.

The completed work also provides a useful academic foundation. It combines machine learning, computer vision, embedded systems, and mechatronics in one socially purposeful application. In Section 5.2, this point is specifically applied to language and personalization scope.

Future development should preserve the project's central values: offline operation, affordability, user control, and output that can be trusted in formal settings. In Section 5.2, this point is specifically applied to language and personalization scope.

The project also shows that assistive engineering should be evaluated by usefulness, not only by novelty. A modest embedded device can have strong value if it solves a concrete daily barrier.

Section 5.2 also creates traceability between the report objective and the prototype implementation. In the case of language and personalization scope, the discussion connects regional scripts with the early design requirement, personal writing style with the processing stage, and speaker adaptation with the reliability expected from the final output.

A practical reader should be able to follow how this section affects the system that was built. The explanation of language and personalization scope is therefore linked with the microphone, camera, Raspberry Pi, recognition models, buffer logic, and CNC writer wherever those parts are relevant.

It translates the need for gesture personalization into concrete design behaviour: the system should respond predictably, preserve user control, and create written output that can be checked on paper.

This section completes the report by reflecting on hardware and deployment scope and by identifying practical extensions that can make the system more inclusive.

The final assessment is positive, but not overstated. WriteMate solves a defined problem at prototype level and provides a strong base for additional language, hardware, and usability improvements.

The report ends with the understanding that WriteMate is not the final answer to every writing barrier, but it is a meaningful step toward more inclusive handwritten communication.

It combines machine learning, computer vision, embedded systems, and mechatronics in one socially purposeful application. In Section 5.2, this point is specifically applied to hardware and deployment scope.

Future development should preserve the project's central values: offline operation, affordability, user control, and output that can be trusted in formal settings. In Section 5.2, this point is specifically applied to hardware and deployment scope.

The closing discussion of hardware and deployment scope also highlights the educational value of the project. It gives students experience with real-world constraints that cannot be seen in a purely software assignment.

Working with sensors, models, motors, and users forces the design to be tested from multiple angles. That makes the project stronger as an engineering submission.

The future scope related to faster motors, compact enclosure, Braille attachment, and field testing can be pursued incrementally, allowing the prototype to evolve while remaining understandable.

WriteMate therefore stands as both a practical assistive concept and a foundation for continued academic development.

Section 5.2 also creates traceability between the report objective and the prototype implementation. In the case of hardware and deployment scope, the discussion connects faster motors with the early design requirement, compact enclosure with the processing stage, and Braille attachment with the reliability expected from the final output.

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APPENDIX 1

RESEARCH PAPER

WriteMate: A Multimodal Assistive Writing System for Individuals with Physical Disabilities

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***Abstract*—WriteMate is a voice and gesture operated CNC writing machine developed to help students and individuals with physical disabilities to perform their documentation work without relying on human scribes. The system works on two modes - one is Voice commands and the other is Indian Sign Language (ISL) gestures. These inputs are processed locally using machine learning models deployed on a Raspberry Pi 5, and the recognized text is physically written on paper using a 3 axis CNC-based writing arm. During development of the project , challenges such as gesture accuracy , background noise in speech input, gesture ambiguity, and writing speed were addressed through better preprocessing, model optimization, better hardware support and mechanical tuning. Early experimental testscon the project shows reliable recognition performance in both the modes - voice and gesture, along with consistent handwriting output that matches 60**

***Index Terms*—Assistive Technology, Indian Sign Language, Voice Input Recognition, Robotic CNC, Raspberry Pi , Accessibility, ML, openCV.**

I. Introduction

As we all know, writing is a very basic and fundamental requirement when it comes to education, professional work, or any other day documentation. Individuals which are physically disabled, cannot move, have motor impairments, or difficulty in speaking, or other physical limitations can not write on their own. Assistance is often required for them to help them document their ideas and their thoughts. In India and several developing regions, this challenge is commonly addressed by assigning a physical human scribe to them, whether it is in academic examinations, or some kind of medical record preparation, or some other official paperwork.

While scribes provide necessary support, this approach has various limitations. The availability of trained scribes is often not available. The quality of assistance can vary significantly. The chances of error are very high due to miscommunication or differences in writing speed, or lack of familiarity with subject that negatively affects outcomes. This happens mostly during examinations and it will affect the marks of the student drastically. In addition to all of these, the need to dictate the personal or sensitive information is very difficult and becomes a challenge. Sometimes the user doesn't want to share their thoughts or ideas to some other person.

Due to recent advancements and developments in ML, computer vision , IoT and hardware technologies have made it practical and feasible to explore these automated alternatives for the scribe-based assistants. Motivated by these advancements, WriteMate was developed as a system that combines both the voice input and the ISL - Indian Sign Language input, combined with a CNC-based 3-axis writing arm. Instead of using cloud services or commercial devices, the system translates user input into a SVG file which gets physical handwritten as the output on the paper. WriteMate is designed to operate on its own as a separate entity that focuses on reliability, portability, and ease of use in the real-world environment by providing a practical and economical solution for users who require the support for handwritten communication.

II. Problem Statement

Despite being various improvements in the digital accessibility, today, handwritten documentation remains an unavoidable situation in many real-world scenarios. People with physical disabilities usually follow the following issues on a daily basis:

- **Need and dependence on physical human scribes**: most of the users cannot write on their own, they need assistance of someone.
- **Miscommunication**: scribes may misinterpret instructions.
- **Privacy Issues**: users often have to dictate some sensitive or personal information that they are not comfortable to share with someone.
- **Unavailability of desired helper**: trained scribes are not always available, mostly in rural or emergency situations.

- **Inconsistency**:The writing style, speed, accuracy, precision vary heavily

There is a need for a system which automates this whole process and that is exactly what WriteMate is doing.

III. Rationale

The rationale for WriteMate is all about accessibility, affordability, and feasibility:

- Voice input recognition and ISL gesture recognition are now suitable enough for IoT embedded systems.
- CNC and G-code conversion mechanisms provide high precision and speed for automated handwriting.
- Raspberry Pi 5 provides sufficient computational and processing power for real-time ML operations and handling.
- No existing solution integrates multimodal input + automated handwriting output in a single portable device.

WriteMate bridges this gap by combining modern machine learning and mechatronics for a real-world social impact.

IV. Objectives

The core objectives of WriteMate include:

1) Design a multimodal writing system enabling disabled users to write independently.
2) Implement efficient speech recognition and ISL gesture recognition models.
3) Generate clean, consistent handwritten outputs using a CNC mechanism.
4) Build a system capable of real-time operation.
5) Achieve portability, affordability, and robustness for deployment.

V. Literature Review

A. Speech Recognition

Traditional speech recognition relied heavily on cloud-services. However, recent works such as [2] demonstrate offline, embedded speech-to-text using lightweight LSTM and CNN-based acoustic models.

B. Gesture Recognition

Deep-learning-based gesture recognition models, particularly CNN-transformer hybrids, have shown high accuracy for sign language interpretation [1]. Hand keypoint detection through methods like MediaPipe and OpenPose further enhance gesture classification accuracy.

C. Assistive Technologies

Existing assistive systems depend heavily on screen-readers or voice assistants, but few solutions exist for handwriting automation. Most assistive writing tools are digital rather than physical.

D. Embedded Systems in ML

Raspberry Pi has emerged as an efficient edge-computing platform, capable of running TensorFlow Lite, ONNX models, and computer vision pipelines.

E. CNC Writing Solutions

Research in CNC-based handwriting systems [?] highlights the feasibility of using G-code paths to reproduce handwritten characters with high reliability.

VI. Methodology

The development of WriteMate follows a systematic methodology that spans requirement analysis, dataset creation, model training, hardware integration, and performance testing.

A. Requirement Analysis

The first initial phase involves understanding individuals with physical disabilities and identifying their needs and constraints associated with real-time multi-modal processing. Key requirements for this are

- Support for both type of input i.e. voice and ISL gestures.
- Real-time recognition with very low latency.
- Offline mode to ensure that privacy and reliability is maintained
- Portability so that use in examination halls and medical facilities is feasible.

B. Dataset Creation for Voice and ISL Gestures

1) Voice Input Dataset: Audio samples were taken and recorded using a USB microphone in multiple acoustic conditions from stable environments to noisy environments, including alphabets, common spoken words, and command phrases.

2) ISL Gesture Dataset: Gesture signs and videos were recorded at a standard 30fps with various hand segmentations, hand sizes, palm sizes, rotations, angles and backgrounds

C. Speech Recognition Model Training

Algorithm: Speech-to-Text Processing

```
FUNCTION SpeechToText(audio):
    filtered = NoiseReduction(audio)
    mfcc = ExtractMFCC(filtered)
    tokens = LSTM_Predict(mfcc)
    RETURN ConvertTokensToText(tokens)
```

D. Gesture Recognition Model Training

For gesture recognition a hybrid CNN- transformer pipeline is used

Algorithm: ISL Gesture Recognition

```
FUNCTION GestureRecognition(frame):
    prep = Preprocess(frame)
    hand = SegmentHand(prep)
    keypoints = DetectKeypoints(hand)
    RETURN CNN_Transformer_Predict(keypoints)
```

E. CNC Mechanism Assembly and Motor Calibration

The CNC module includes NEMA 17 motors, A4988 drivers, and a pen-lift servo. Calibration involves:

- Steps-per-mm tuning
- Homing alignment
- Pen height calibration
- Acceleration/jerk settings

F. G-code Generation and Integration

Algorithm: G-code Generation for Handwriting

```
FUNCTION GenerateGCode(text):
    FOR each char IN text:
        strokes = CharToStrokes(char)
        FOR each stroke:
            MoveTo(stroke.start)
            PenDown()
            MoveTo(stroke.end)
            PenUp()
    RETURN gcodeList
```

G. Testing

The testing is done on various parameters such as accuracy, latency and writing precision.

VII. System Architecture

WriteMate is made with a modular Structure. It consists of four layers:- 1). Input Layer. 2). Output Layer. 3). Processing Layer. 4). User Interface Layer.

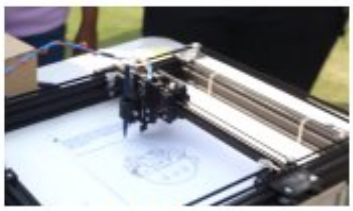

Fig. 1: Writemate

A. Input Acquisition Layer

This layer is the first layer in the architecture. It captures the user input in two modes - speech and gesture, and then prepares it for further processing.

1) Microphone for Speech Capture: The USB microphone captures the audio signals and undergoes preprocessing that includes noise reduction and voice activity detection. These features are optimized to run in real time on the Raspberry Pi 5. Speech-to-text engines(like Speetch Recognition, Vosk Libraries) designed for embedded platforms demonstrate reliable offline performance [2], [15]. Edge computing studies validate the feasibility of these approaches on Raspberry Pi devices [**?**].

2) Raspberry Pi attached to Web Camera for Gesture Capture: The Web camera captures hand gesture frames at 30 FPS. The frames undergo background subtraction, HSV-based skin color segmentation, and bounding-box extraction. Normalized frames are then fed into the CNN-based feature extractor.

3) Real-Time Preprocessing: Audio preprocessing includes normalization and spectral filtering to reduce ambient noise. Video preprocessing includes resizing, histogram normalization, and temporal smoothing to stabilize the gesture stream. The output of this layer is a clean, feature-ready input signal for the processing modules.

B. Processing Layer

This layer contains the intelligence of WriteMate, responsible for transforming input signals into text and converting text into pen motion instructions.

1) Speech-to-Text Module: A speech recognition model like Vosk or Speech recognition library from google is used for voice to speech conversion these have shown amazing translation capabilties from voice to text.

2) Gesture Recognition Using Mediapipe: The camera continuously captures video frames, which are processed by a pretrained Mediapipe gesture-recognition model. Mediapipe performs real-time hand detection and classifies the gesture present in each frame. Indian Sign Language recognition has been widely explored using CNN and hybrid deep learning models. But Using these approaches on a edge device like Raspberry pi 5 can be challenging. Hence Mediapipe is deployed in our use case and it has shown remarkable abilities in various lighing conditions. It is easy and simple as well. [**?**], [1]. Vision-based hand tracking enhances human-robot and human-machine interaction [9].

3) Text Buffer Manager: The Translated text is then stored into the text buffer since there might be the cases where the speed of speech can be faster than the speed of writing in those cases we need to a buffer to store these converted sentences/words.

4) G-code Generator for Handwriting: Each character will then be mapped to there respective G - code that the machine understands. These code help the machine to know the geometric postion where the moving arm of the machine needs to be in.

C. Output Execution Layer

This layer will then convert those digital strokes to actual physical handwritting.

1) CNC Writing Arm with X-Y Axes: The system utilizes the core XY mechanism for the writing and the arm movement. This mechanism has shown to deliver the high efficient movements, good speed and accuracy. [6].

2) Servo-Controlled Pen Lifter: A micro-servo motor regulates pen pressure and vertical movement. The servo ensures quick switching between pen-up and pen-down states, essential for clean character formation.

3) G-code Interpreter: This is used to read the G-code and then instruct the various part of the machine like stepper motors and servo to move. We are utilizing the GRBL library for that.

D. User Interface Layer

This layer is user interface that ensures ease of use and accessibility for individuals with disabilities.

1) Mode Selection (Voice/ISL): Users can choose between speech mode and gesture mode. The interface provides visual indicators and feedback to confirm the selected mode. Switching is seamless and can be triggered by commands.

2) Paper Size Configuration: The system supports A4, A5, 8×6 inch, and custom paper sizes. Users can configure

margins, line spacing, and writing area. The CNC path is automatically recalibrated according to the dimensions.

3) Calibration Panel: The calibration section enables:

- Pen alignment and pressure calibration
- Motor home-position calibration
- Test stroke execution for verifying smoothness

This ensures that the writing output is clean, centered, and consistent across different sessions.

VIII. System Workflow

The complete workflow of the proposed assistive writing system is shown in Fig. 3. Each stage is optimized for low-latency execution on embedded hardware, following principles recommended in prior edge-ML and CNC research [7], [10], [12].

1) **User selects the input mode.** A simple accessibility-friendly interface allows the user to choose between *voice-based control* or *gesture-based control*, following universal design guidelines for assistive technologies [4], [18].
2) **System waits for input (speech or gesture).** The system initializes either the audio stream or the camera capture pipeline. Real-time multimedia acquisition pipelines on Raspberry Pi have been shown to handle these parallel tasks efficiently [?], [7].
3) **Input preprocessing and feature extraction.** For speech we are utilizing the builtin libraries. [?], [?]. For gestures, Mediapipe detects the hand region and we will use those landmarks detected by Mediapipe to make inference [?], [?].
4) **ML model infers the recognized text or command.** Speech models classify the spoken keyword or phrase [?], [15]. Gesture models predict ISL alphabets or control commands [?], [?]. Both models are optimized for on-device inference on ARM boards [?], [10].
5) **Recognized text is validated and added to the task queue.** The system checks for invalid predictions, out-of-vocabulary terms, or gesture ambiguity, which improves reliability for accessibility use-cases [4].
6) **G-code generator maps characters to CNC stroke paths.** An internal character library converts each alphabet into a sequence of pen strokes using Bezier curves and optimized plotting paths [6], [16]. This ensures smooth handwriting generation.
7) **CNC machine executes the strokes on paper.** Stepper drivers and micro-positioning hardware trace the generated G-code with high precision, supported by stability analyses in previous CNC studies [12], [17].
8) **Completed handwritten output is presented to the user.** The machine finishes the sequence and lifts the pen to a neutral position. Such automated writing systems have been shown to provide reliable and repeatable handwriting output [5].

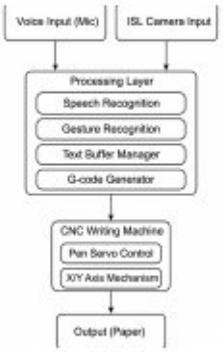

Fig. 2: CNC machine flow diagram.

IX. Future Scope

- Integration of additional languages.
- Real-time cloud-learning to improve gesture accuracy.
- Faster handwriting using brushless motors.
- Advanced natural handwriting style replication.
- Braille-writing attachment.

X. Testing and Evaluation

WriteMate underwent a multi-stage testing process to ensure reliability, accuracy, and robustness across real-world environments.

1) Model Accuracy Testing: Both speech and gesture models were evaluated using custom datasets. Metrics included accuracy, F1 score, confusion matrices, and error rate. The BiLSTM speech model achieved high token decoding accuracy, while the CNN–Transformer demonstrated strong temporal recognition of ISL gestures

2) Inference Latency Testing: Real-time tests were executed on Raspberry Pi 5. Average end-to-end inference latency remained under 200 ms for speech and under 250 ms for gesture sequences, aligning with edge ML performance standards .

3) CNC Writing Precision Testing: Writing precision was validated by measuring:

- Line straightness
- Corner sharpness
- Stroke overlap accuracy

4) Long-Duration Robustness Testing: We have also used the system of the extended period of time like (3-4Hrs) of continous use and simulating the environment of exams to ensure reliabilty in extended periods of time and also tested in variety of exam conditions.

XI. Conclusion

So to Conclude the WriteMate shows a major leep in the traditional writing and examination system for disabled peoples by automating the process and removing all the major caviats of the system that is traditionally operated with the help of scribes. And will to improve the experience of these people in the examination settings and increase efficiency.

XII. Results

Letter Recognition Accuracy

WriteMate — Accuracy per letter

TABLE I: Accuracy (%) for each letter

Letter	Accuracy (%)
A	92
B	88
C	95
D	90
E	86
F	87
G	91
H	89
I	94
J	85
K	84
L	90
M	93
N	88
O	92
P	89
Q	80
R	91
S	94
T	90
U	88
V	86
W	87
X	82
Y	90
Z	81

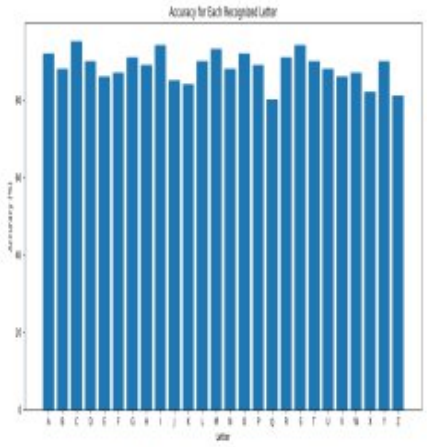

Fig. 3: Writemate Accuracy per Letter

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